Diploma in Information and Communication Technology

Human Computer Interaction

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CIVICLINK SA

PART A: UNDERSTANDING THE PROBLEM AND USERS

A1: PLATFORM IDENTIFICATION AND EVALUATION:



CityMenderSA is a South African digital service delivery reporting system designed to enable citizens to report and track infrastructure-related issues such as potholes, water leaks, sanitation failures, and electricity faults.

The platform provides features such as real-time issue tracking, interactive mapping, and community-driven updates, aiming to improve transparency, accountability, and communication between residents and municipalities.

GOOD DESIGN FEATURES:

- **Interactive Map-Based Reporting:**

  The platform allows users to report issues by placing a pin on a map. This enhances **effectiveness**, as users can accurately specify locations, and supports the principle of **visibility**, making system status and location information clearly perceivable

- **Real-time Issue tracking with Reference Numbers**

  Each report is assigned a unique reference number for tracking progress. This supports **memorability** and **efficiency**, as users can easily revisit and monitor reports, while also demonstrating **feedback**, by informing users of system status.

- **No Registration Required**
  Users can report issues without creating an account. This improves **learnability** and **efficiency**, reducing barriers to entry and supporting inclusive interaction design by accommodating users with low digital literacy

- **Offline Reporting Capability**
  The system allows users to log issues without internet access and sync later. This supports **effectiveness** in low-connectivity environments (common in rural Eastern Cape) and aligns with **accessibility and inclusiveness**, key principles in user-centred design.

- **Community Based Updates (Collaborative Interaction)**

  Users can update and validate reports collaboratively. This enhances **user experience goals** such as **engagement and satisfaction**, while also supporting **feedback** and **consistency** in shared system interaction

POOR DESIGN FEATURES:

- **Lack of Direct Municipal Integration:**

  The platform operates independently from government systems. This reduces effectiveness, as reported issues may not always lead to action, and violates feedback, since users may not receive official resolution updates.

- **Potential Information Overload on Map Information:**

  Displaying many issues simultaneously may overwhelm users. This negatively impacts **learnability** and violates the principle of **visibility**, as important information may become difficult to interpret.

- **Limited Structured Feedback after submission:**
  Although reports are tracked, feedback may depend on community updates rather than system-generated responses.
  This weakens **feedback** and reduces **user confidence**, negatively affecting the overall user experience.

- **Over Reliance on User Participation:**
  The system depends heavily on users to update reports.
  This impacts **efficiency** and **reliability**, as inactive users may lead to outdated or incomplete information, violating **consistency**.

- **Advanced Features May Reduce Learnability:**
  Features such as augmented reality reporting and AI-based analysis may be complex for some users.
  This reduces **learnability** and may create usability barriers, particularly for elderly or low-tech users, conflicting with **accessibility principles**.

A2: USER ENGAGEMENT

To better understand the interaction challenges identified in CityMenderSA, two users from a local South African community in Eastern Cape context were engaged. The purpose of this engagement was to explore how different user groups experience service delivery reporting systems, particularly in relation to usability, feedback, and accessibility.

PERSONA 1: Nomsa Ngubengcuka



Age: 59

Location: Eastern Cape, King William's Town, Zalarha

Occupation: Community Caregiver

Digital Skills: Low level

Frustrations: Confused by technical terms and icons

 Does not receive clear feedback after reporting issues

Feels ignored by municipality systems

Nomsa notices a burst water pipe near her home in a rural Eastern Cape village. She attempts to report the issue using CityMenderSA. However, she struggles to understand the map interface and is unsure where to place the location pin. After finally submitting the report, she receives no clear confirmation or feedback that the issue has been logged successfully.

Days later, she is still unsure whether the municipality has acknowledged the issue, leading to frustration and repeated informal complaints.

PERSONA 2: SIPHO GABELANI



Age: 25

Location: Eastern Cape, East London, Mdantsane

Occupation: University Student

Digital Skills: Digitally Skilled Youth

Frustrations: Lack of **system feedback and status updates**

 Poor **information visibility** due to cluttered interface

 Reduced **efficiency** in tracking issues

 Weak **consistency** in municipal response data

Sipho reports multiple potholes on a busy road using the platform. He uses the map feature easily and submits several reports. However, over time, he notices that most issues remain "unresolved" without updates. The map becomes overcrowded with unresolved pins, making it difficult to identify new issues.

He also realizes that municipal response is inconsistent, reducing his trust in the system.

PROBLEM STATEMENT

Residents in South African communities, particularly in the Eastern Cape, experience significant usability and user experience challenges when using digital service delivery reporting systems due to poor feedback mechanisms, complex interface structures, and lack of transparent issue tracking. These limitations reduce system trust, decrease user engagement, and hinder effective communication between citizens and municipalities.

WHAT MAKES AN INTERACTIVE SYSTEM EFFECTIVE?

An interactive system is one that:

- Achieves **usability goals** such as efficiency, learnability, and effectiveness.
- Provides continuous **feedback** to keep users informed of system status.
- Supports **user-centred design**, ensuring accessibility for diverse user groups.
- Delivers a positive **user experience (UX)** characterized by trust, satisfaction, and engagement.
- Aligns with real-world user contexts, especially in low resource environments.

These principles are essential in ensuring that systems are not only functional but also meaningful and usable in everyday life.

HOW WILL THE PROPOSED SYSTEM IMPROVE USER EXPERIENCE?

The proposed redesigned system **CivicLink SA** will improve user experience by:

- Introducing clear confirmation and progress feedback after issue submission
- Simplifying the interface to improve **learnability for low-literacy users**
- Providing **structured tracking updates** to improve transparency and trust
- Ensuring inclusive design that supports both rural and urban users

This directly addresses the usability and UX failures identified in CityMenderSA and aligns with **user-centred interaction design principles**.

The evaluation of CityMenderSA revealed key usability challenges, particularly in feedback, visibility, and system transparency. These findings informed user research, which highlighted significant differences in user needs between rural and urban users.

The defined problem therefore becomes the foundation for Part B, where **CivicLink SA**, a redesigned interactive system, will be developed to improve usability, enhance feedback mechanisms, and support inclusive interaction for all users in the Eastern Cape context.

PART B: CONCEPTUAL DESIGN AND INTEACTION THINKING

CIVICLINK SA

1. CORE FUNCTIONALITIES OF CIVICLINK SA

The core functionalities of CivicLink S.A. are deliberately designed to resolve the specific usability and user experience breakdowns identified in CityMenderSA, particularly those related to poor feedback, interface complexity, information overload, and lack of transparency.

FUNCTIONALITY	JUSTIFICATION
Guided Reporting with Simplified Input	Addresses Nomsa's difficulty with complex map-based interaction and unclear interface elements. Improves **learnability and effectiveness** by using clear affordances and structured steps.
Immediate Confirmation and Continuous System Feedback	Resolves the lack of feedback identified in A1 and experienced by both Nomsa and Sipho. Enhances **feedback, user trust, and satisfaction**
Structured Issue Progress Tracking	Addresses lack of transparency and unclear issue status. Improves **visibility and memorability**, enabling users to understand system progress easily.
System-Generated Updates (Reduced User Dependency)	Addresses over-reliance on user participation. Improves **consistency, reliability, and effectiveness** by ensuring accurate and continuous updates.

2. INTERACTION TYPES

INTERACTION TYPE	JUSTIFICATION
Instructing	Supports guided reporting and structured task completion. Addresses Nomsa's struggle with complex interactions by improving learnability and efficiency
Manipulating and Navigating (Controlled)	Enables users to interact with a simplified map interface. Resolves Sipho's issue with clutter and improves **visibility and efficiency**
Exploring (Structured Exploration)	Allows users to track issue progress and view updates. Addresses lack of transparency and improves **engagement, trust, and satisfaction**.

3. INTERFACE METAPHOR

CivicLink S.A. is based on a **journey-based interface metaphor**, where each reported issue is represented as a "journey" progressing through different stages (e.g., submission, verification, resolution).

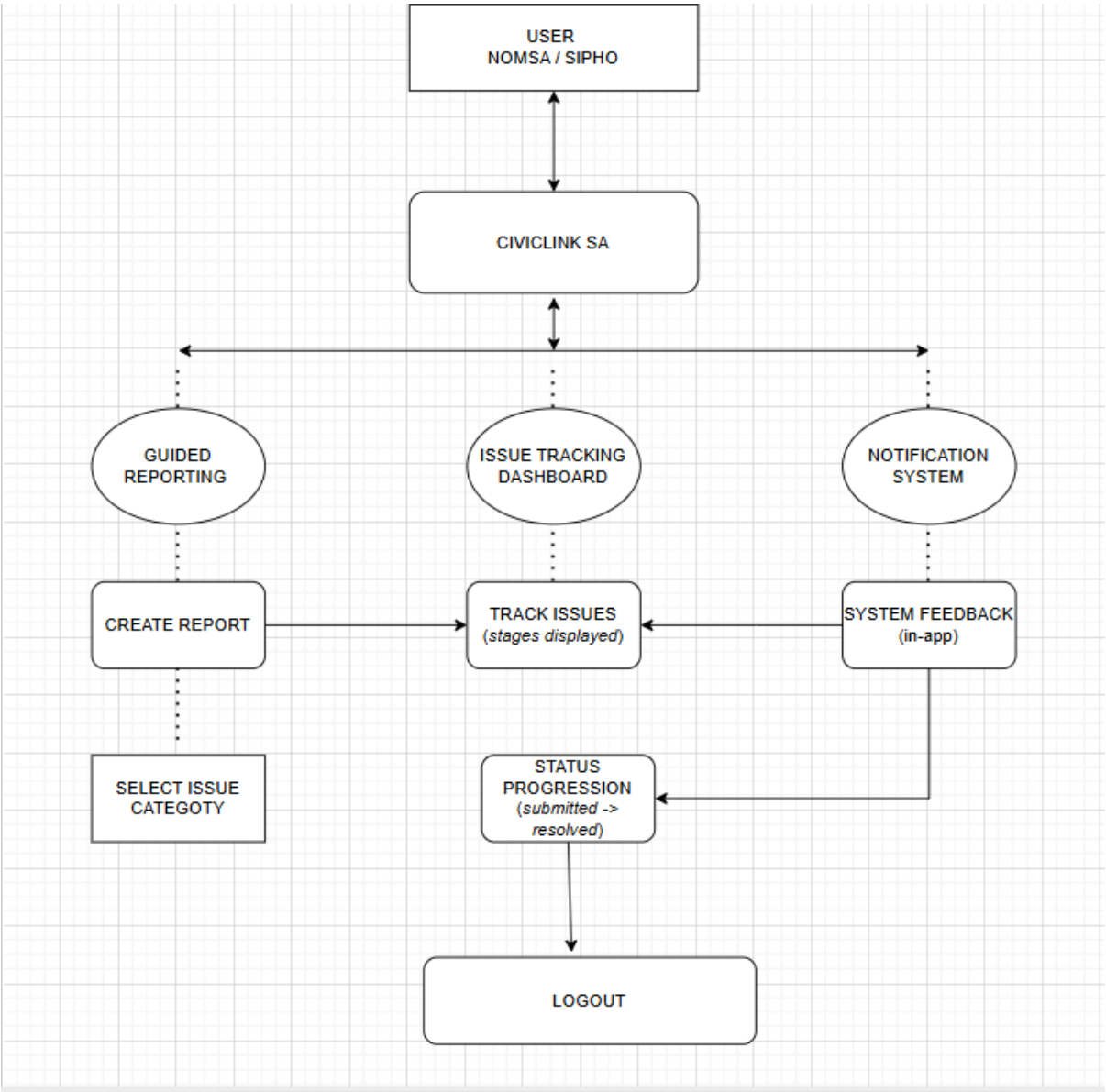
Users perceive each issue as moving through a **clear, structured lifecycle**, like tracking a parcel delivery or service request.

*This metaphor was selected to address the **lack of transparency and feedback** identified in Part A.*

- *It enhances **visibility**, as users can clearly see system status.*
- *It improves **memorability**, as users easily understand progress stages.*
- *It strengthens **user trust and satisfaction**, as progress is continuously communicated.*
- *It aligns with **real-world mental models**, making the system intuitive for both low and high literacy users.*

For Nomsa, the metaphor simplifies understanding of system processes, while for Sipho, it provides structured and efficient tracking.

CONCEPTUAL MODEL DIAGRAM



Narrative explanation:

The conceptual model of CivicLink S.A. is designed as a structured service reporting and tracking system, where users interact with clearly defined system components to report, track, and receive updates on service delivery issues.

The model is based on a combination of instructing, manipulating, and exploring interaction types, enabling users to perform purposeful actions while maintaining clarity and control over system process.

Linkage to A1- A2- PROBLEM – DESIGN

The evaluation of CITYMENDER SA and user research revealed the following issues:

PLATFORM EVALUATION (CITYMENDER SA)	USER RESEARCH (NOMSA & SIPHO)
Lack of feedback	Nomsa struggles with complex interface and lack of guidance
Information overload	Sipho experienced cluttered displays and poor tracking visibility
Complex interaction structures	
Lack of transparency	

Conceptual Model (Translation to design)

The conceptual model resolves these issues through:

- **Guided Reporting Component** → simplifies interaction (Nomsa)
- **Tracking Dashboard** → improves transparency (Sipho)

SYSTEM BEHAVIOUR

The system operates as follows:

1. The user initiates interaction through the guided reporting module
2. The system constrains input through step-by-step actions (reducing complexity)
3. A report is created and stored within the system.
4. The system assigns a structured status progression.
5. The user tracks the issue through a dashboard interface.

USABILITY GOALS

Learnability:

- The conceptual model improves learnability through:
  - Guided reporting steps
  - Clear system structure
  - Reduced reliance on complex map interaction

Directly addresses Nomsa's difficulty understanding the system.

Efficiency

- Efficiency is improved through:
  - Structured tracking dashboard
  - Filtered map interface.
  - Reduced need for repeated reporting

Addresses Sipho's frustration with inefficient tracking

USER EXPERIENCE GOALS

Trust

- Trust is enhanced through:
  - Continuous system feedback
  - Transparent issue tracking

Resolves the lack of confidence in the original system.

Satisfaction

- User satisfaction is improved by:
  - Clear confirmation messages
  - Predictable system behaviour
  - Reduced frustration

Ensures a positive interaction experience for both personas.

HOW DOES THE DESIGN IMPROVES THE EXISTING PLAFORM (CITYMENDER SA)

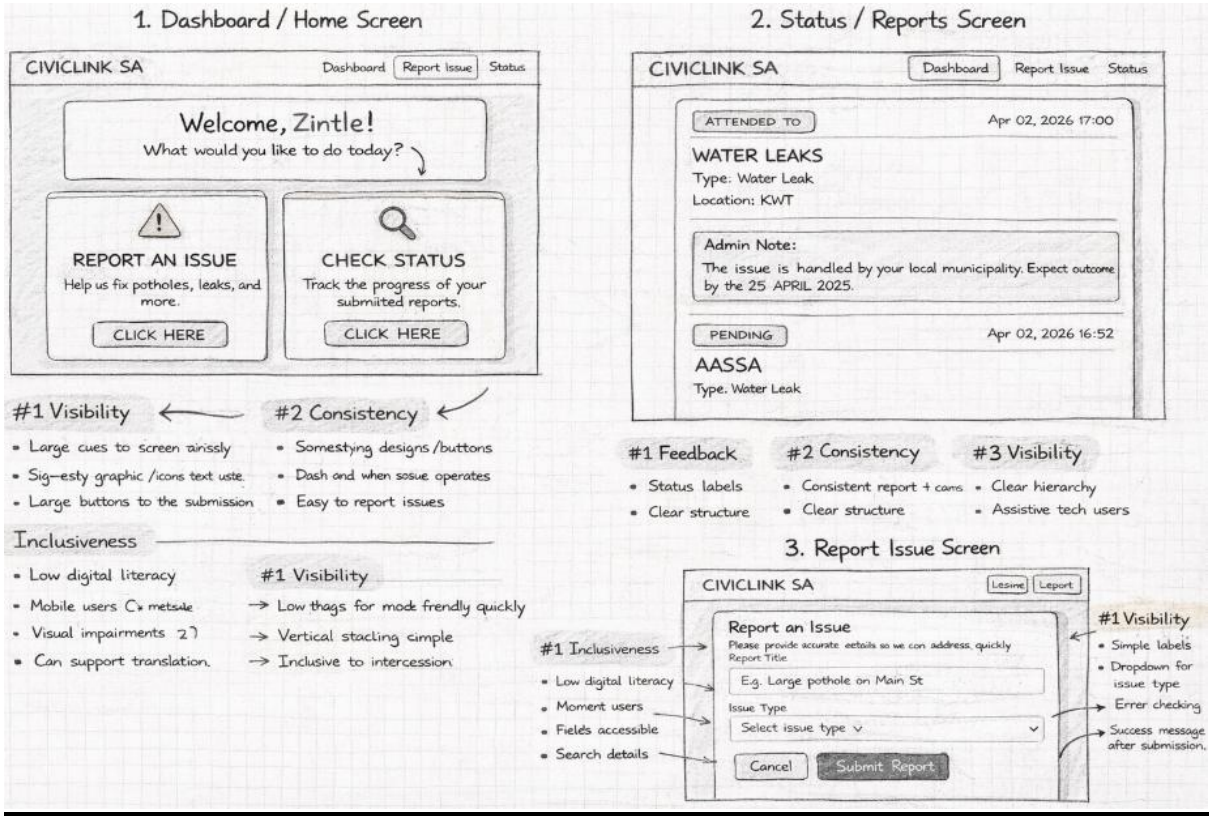
PROBLEM (CITYMENDER SA)	CONCEPTUAL MODEL SOLUTION
Lack of feedback	Notification system provides continues updates
Complex interface	Guided reporting simplifies interaction
Over-reliance on users	System – generate updates ensure consistency
Lack of transparency	Structures staging improve visibility
Information overload	Filtered Map reduces clutters

SUPPORTS FOR PERSONAS

NOMSA	SIPHO
Filtered and controlled map display	System-generated updates
Immediate confirmation and continuous feedback	Structured issue progress tracking
Step-by-step instructions and clear affordances	Notification system (in-app)

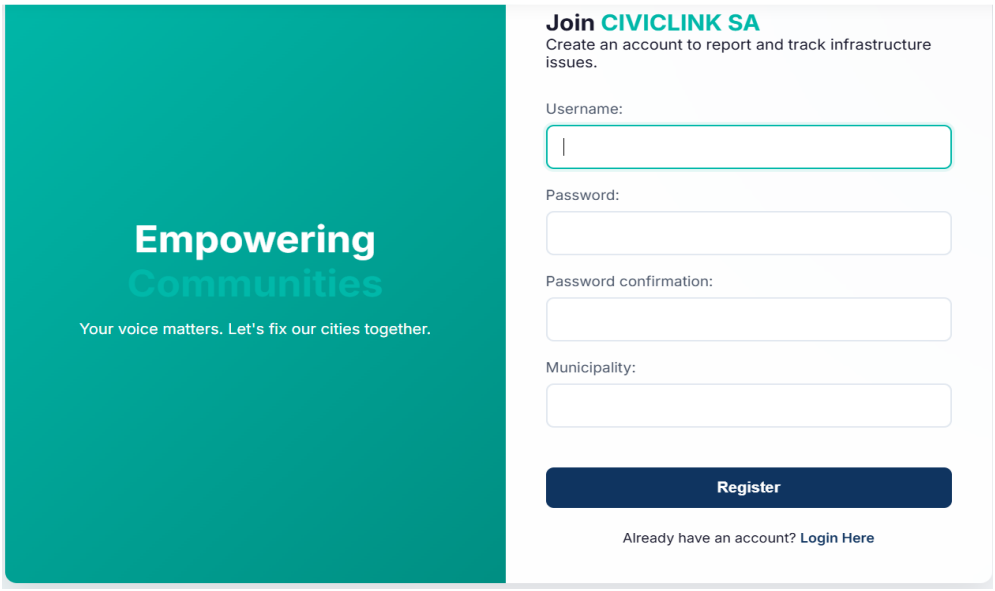
PART C: INTERFACE REPRESENTATION AND PROTOTYPING

INTERFACE SKETCHES:



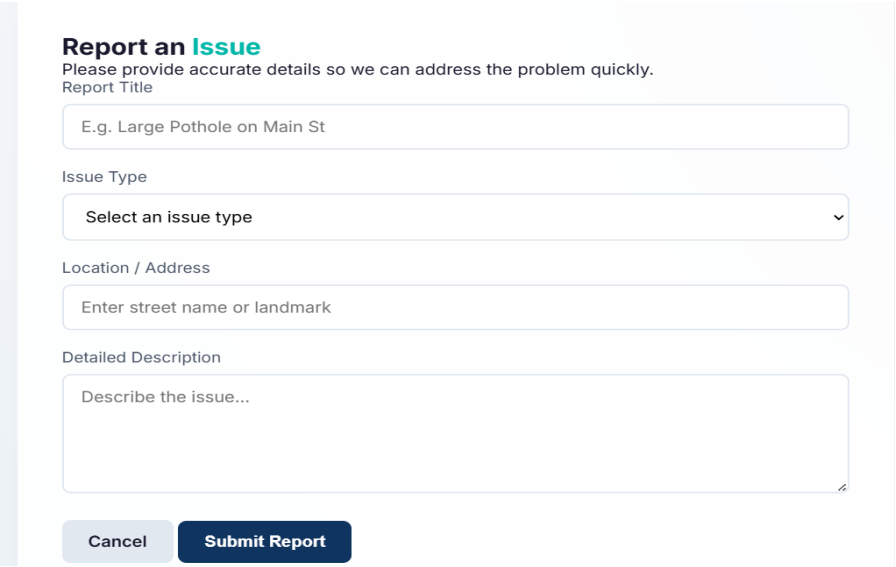
DJANGO WIREFRAME:

GITHUB LINK: https://github.com/AsavelaTshonisile/CIVICLINK-SA.git



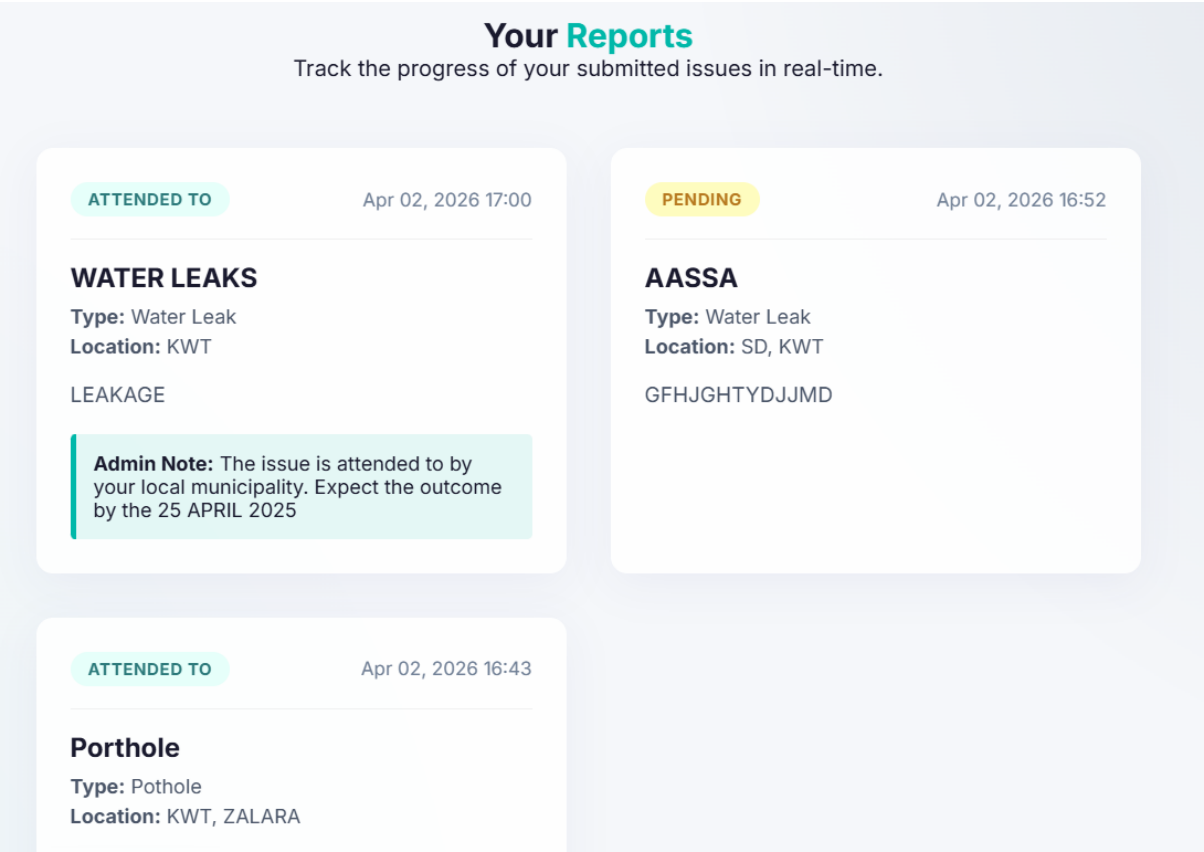
This is the entry point for new users, featuring a clean, split-screen design.

- Left Side (Visual/Brand): A large teal block with the slogan *"Empowering Communities"* and a call to action.
- Right Side (Form): A vertical stack of input fields for account creation:
    - Fields: Username, Password, Password Confirmation, and Municipality.
    - Action: A prominent dark blue Register button at the bottom.
- Top Header: Contains the platform logo and "Login" or "Sign Up" links in the top right corner.



This is a functional form designed for data entry, using a single-column layout.

- Header: Clear title ("Report an Issue") with a brief instruction sub-text.
- Form Fields: * Report Title: A text box for a brief summary (e.g., "Large Pothole").
    - Issue Type: A dropdown menu to categorize the problem.
    - Location / Address: A text box for street names or landmarks.
    - Detailed Description: A larger text area for specific notes.
- Footer Actions: Two distinct buttons: a light grey Cancel button and a dark blue Submit Report button to finalize the entry.



This page acts as a dashboard for monitoring previously submitted issues.

- **Grid Layout:** Information is organized into individual white "cards" for each report.
- **Card Structure:**
    - **Status Badges:** Color-coded labels in the top left of each card (e.g., "ATTENDED TO" in teal or "PENDING" in yellow).
    - **Timestamps:** The date and time of submission are displayed in the top right.
    - **Content:** The title (e.g., "WATER LEAKS"), type, location, and a brief snippet of the user's description.
    - **Feedback Loop:** Some cards include a highlighted **Admin Note** section at the bottom to provide direct updates from the municipality.

PART D: AI SUPPORTED DESIGN IMPROVEMENT

AI FEEDBACK SUMMARY:

I used an AI tool to evaluate my initial sketches and wireframes. The AI identified the following usability issues based on UX principles:

- **Data Accuracy Issues**: The "Location" field is a plain text box, which may lead to vague addresses like "near the shop," making it hard for the city to find the problem.
- **Minimal Evidence**: There is no option to upload a photo, which is essential for verifying the severity of a "Water Leak" or "Pothole."

AI GENERATED SUGGESTIONS FOR IMPROVEMENT

The AI provided the following three suggestions to enhance the user experience:

1. **GPS Integration**: Add a "Use Current Location" button to ensure coordinate accuracy.
2. **Media Upload**: Include an "Attach Photo" button in the reporting form to provide visual proof.

DESIGN REFINEMENT AND JUSTIFICATION

SUGGESTION	ACTION TAKEN	JUSTIFICATION
GPS button	Accepted	It reduces user effort (Human-Computer Interaction) and prevents human error in typing addresses.
Attach Photo	Accepted	This aligns with the "Problem" in Part A; photos help the municipality prioritize urgent repairs quick



PART E: REFLECTION

The design process for CivicLink SA has been a transformative exercise in shifting from a developer-centric mindset to a user-centred design (UCD) approach. I learned that UCD is not just about making an interface look "nice," but about ensuring the system speaks the user's language and minimizes their cognitive load. Applying interaction design principles, such as visibility and feedback, taught me that a user should never have to guess if their report was successful; the system must provide immediate validation.

Evaluating an existing platform before building my own was a turning point. It allowed me to identify "pain points"—such as cluttered navigation or hidden status updates—that I might have otherwise duplicated. This "competitor analysis" highlighted that simplicity is often more difficult to achieve than complexity, but far more rewarding for the end-user.

I faced several challenges during this project. Understanding users was difficult because I had to move past my own assumptions and consider a diverse South African demographic with varying levels of digital literacy. Defining the problem required narrowing down a broad "city issue" into a specific, actionable reporting tool. Translating ideas into interfaces was perhaps the hardest step; it is easy to imagine a feature like "live tracking," but difficult to design it in a way that feels intuitive on a small mobile screen.

My understanding of good vs. poor design has evolved significantly. I now realize that poor design creates barriers (like a location field that is hard to fill out), while good design removes them (like a GPS button). This focus on User Experience (UX) influenced my final solution to prioritize speed and transparency, ensuring that citizens feel heard and municipalities can act efficiently.

6. AI Reflection Requirement

AI served as a critical design assistant by acting as a "second pair of eyes." It helped identify "blind spots" in my original wireframes.

The strengths of using AI include its ability to provide instant, objective critiques based on established HCI heuristics. However, its limitations are evident in its lack of local context; it might suggest "Social Logins" without realizing that municipal systems often require specific, secure residency data. I ensured the work remained my own by critically filtering the AI's advice—only accepting changes that fit the South African municipal context and manually coding the Django elements to ensure I understood the underlying logic.

7. AI Usage Declaration (100–150 words)

I declare that I utilized Gemini (an AI language model) to assist in the completion of this assignment. AI was specifically used in Part C to evaluate my initial UI sketches against usability heuristics and to generate suggestions for layout improvements (such as adding GPS and photo upload features). Additionally, AI helped structure the Reflection in Part E and provided code snippets for Bootstrap implementation in the Django wireframe.

To ensure originality and understanding, I did not copy-paste AI responses blindly. Every suggestion made by the AI was evaluated against the project requirements. I manually integrated the suggested UI components into my design and wrote the justifications based on my own learning of the course material. The final prototype reflects my own creative decisions and technical implementation.
